

Exact Rank Two for the Extended Oscillation Rank

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Status

Partial result, likely valid; human verification recommended. The original question asks for every nonzero countable ordinal. This packet proves the exact value 2 for every allowed Baire level $\xi \geq 2$; it does not settle arbitrary successor ordinals.

Source Question

Leung, Ng, and Tang ask the following after Proposition 3.6 on page 11 of their paper [1].

Remark. In an uncountable Polish space (X, τ) , any nonzero countable ordinal is in the range of $\beta = \beta_1^*$. Propositions 3.4 and 3.6 say that this is “almost” true for $\xi \geq 2$. It would be interesting to find out if it is true exactly.

Question. Let (X, τ) be an uncountable Polish space and let ξ be a countable ordinal ≥ 2 . Is it true that for any nonzero countable ordinal ζ , there exists $f \in \mathfrak{B}_\xi(\tau)$ such that $\beta_\xi^*(f) = \zeta$?

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The source proves that every nonzero countable limit ordinal occurs and that, for every nonzero ζ , some characteristic function has rank strictly between ζ and $\zeta + 2$ [1, Propositions 3.4 and 3.6].

Definitions

For $f: X \rightarrow \mathbb{R}$, a closed set F , and $\varepsilon > 0$, let $D_\sigma^1(f, \varepsilon, F)$ be the set of points $x \in F$ such that every σ -neighborhood of x contains $x_1, x_2 \in F$ with $|f(x_1) - f(x_2)| \geq \varepsilon$. Iterate this derivation transfinitely. The oscillation rank $\beta_\sigma(f)$ is the supremum over $\varepsilon > 0$ of the first stage at which the iterated derivative is empty.

If $f \in \mathfrak{B}_\xi(X, \tau)$, the extended rank is

$$\beta_\xi^*(f) = \min\{\beta_\sigma(f) : \tau \subseteq \sigma \subseteq \Sigma_\xi^0(\tau), \sigma \text{ Polish, and } f \in \mathfrak{B}_1(X, \sigma)\}.$$

Proof Intuition

Rank one is exactly potential continuity under an admissible refinement. For a characteristic function, potential continuity would make its support clopen. Since every open set of an admissible topology is already Σ_ξ^0 in the original topology, rank one is possible only for supports in Δ_ξ^0 .

Strictness of the Borel hierarchy supplies a support $A \in \Pi_\xi^0 \setminus \Sigma_\xi^0$. It cannot become clopen, so rank one is excluded. On the other hand, its complement is Σ_ξ^0 , and the topology-refinement lemma from the source paper can make that complement open, hence make A closed. The indicator of a closed set has rank at most two: after one oscillation derivative only points of A remain, and the function is constant there.

Theorem 1 (Exact value two). *Let (X, τ) be an uncountable Polish space and let $2 \leq \xi < \omega_1$. There exists $f \in \mathfrak{B}_\xi(X, \tau)$ such that*

$$\beta_\xi^*(f) = 2.$$

The function f may be chosen to be characteristic.

Proof. By strictness of the Borel hierarchy on uncountable Polish spaces [3, Section 22], choose

$$A \in \Pi_\xi^0(\tau) \setminus \Sigma_\xi^0(\tau).$$

In particular $A \in \Delta_{\xi+1}^0(\tau)$, so the Lebesgue–Hausdorff characterization gives $f = \chi_A \in \mathfrak{B}_\xi(X, \tau)$.

Set $G = X \setminus A \in \Sigma_\xi^0(\tau)$. Apply the topology-refinement lemma [1, Lemma 3.1] to G and the original topology. It gives a Polish topology σ such that

$$\tau \subseteq \sigma \subseteq \Sigma_\xi^0(\tau) \quad \text{and} \quad G \in \sigma.$$

Thus A is σ -closed. The indicator of a closed set is Baire class one in a metrizable topology, so σ is admissible for the minimum defining $\beta_\xi^*(f)$.

We claim that $\beta_\sigma(f) \leq 2$. If $\varepsilon > 1$, the first derivative is empty. If $0 < \varepsilon \leq 1$, every point of the open set G has a σ -neighborhood on which f is constantly zero. Hence

$$D_\sigma^1(f, \varepsilon, X) \subseteq A.$$

On this derived closed set f is constantly one, so

$$D_\sigma^2(f, \varepsilon, X) = \emptyset.$$

Therefore $\beta_\xi^*(f) \leq \beta_\sigma(f) \leq 2$.

It remains to exclude rank one. Suppose $\beta_\xi^*(f) = 1$. Then an admissible Polish topology ρ satisfies $\beta_\rho(f) = 1$. Vanishing of every first oscillation derivative is equivalent to continuity of f . Consequently

$$A = f^{-1}((1/2, \infty)) \quad \text{and} \quad X \setminus A = f^{-1}((-\infty, 1/2))$$

are both ρ -open. Since $\rho \subseteq \Sigma_\xi^0(\tau)$, both sets belong to $\Sigma_\xi^0(\tau)$, and therefore $A \in \Delta_\xi^0(\tau)$. This contradicts $A \notin \Sigma_\xi^0(\tau)$. Thus $\beta_\xi^*(f) \geq 2$, completing the proof. $\square$

Verification Notes

Step	Verdict	Adversarial check
Existence of A	External, standard	The Borel hierarchy is strict at every countable level on every uncountable Polish space.
$\chi_A \in \mathfrak{B}_\xi$	Valid	Both fibers are in $\Delta_{\xi+1}^0$; the indicator therefore has Baire class at most ξ .
Admissible refinement	Valid	Lemma 3.1 applies to the single set $G = A^c \in \Sigma_\xi^0$ and preserves the required upper bound on the topology.
Upper rank bound	Valid	Closedness of A removes its complement at the first derivative; constancy on the remaining set removes it at the second.
Lower rank bound	Valid	Rank one forces continuity, hence both fibers are open in an admissible topology and therefore both are originally Σ_ξ^0 .
Match to source	Valid, partial	The source asks for all nonzero ordinals; only the exact value 2 is proved here.

No finite computation is relevant. The main human-review focus should be the use of strict Borel hierarchy on arbitrary uncountable Polish spaces and the precise convention that $\mathfrak{B}_\xi$ contains functions of lower Baire class.

Focused Full-Upgrade Pushes

Three routes toward all successor ordinals were tested.

1. A one-point accumulation gadget can add one ordinary oscillation derivative, but it is not refinement-resistant: a singleton is Δ_2^0 and can be made clopen by an admissible refinement when $\xi \geq 2$.
2. The Hausdorff difference hierarchy over Π_ξ^0 gives a natural lower bound, but the available comparison is $\alpha_\xi(f) \leq 2\alpha_\xi^*(f)$. Without a construction attaining the sharp factor-two calibration and a matching upper topology, it does not force a prescribed successor value of β_ξ^* .
3. Nested Π_ξ^0 supports and multilevel characteristic sums give short upper derivative chains after refinement. No lower bound was found that prevents a different admissible refinement from collapsing those chains. Thus this route was not promoted beyond the exact-rank-two case.

Scope and Novelty Check

Combining the theorem with the constant-function case and [1, Proposition 3.6], the known exact range contains 1, 2, and every nonzero countable limit ordinal. Arbitrary successors, beginning with 3, remain unresolved by this packet.

The bounded search covered the run’s registry, solution, attempt, and proof-gap indexes; the full sources of arXiv:1701.05649 and arXiv:1406.5724; exact source-question phrases; searches for the extended rank and its range; and citation metadata for the source article. No explicit prior statement of this exact-rank-two subcase or later full answer was located. The search is not exhaustive, so the novelty assessment is moderate rather than definitive.

Human Review Recommendation

Send for human review as a short, checkable partial result. A reviewer should confirm the two standard descriptive-set-theoretic inputs and then decide whether the rank-two observation is sufficiently nontrivial for separate publication or is best retained as run memory. Do not treat this packet as a solution of the full range problem.

References

- [1] D. H. Leung, H.-W. Ng, and W.-K. Tang, *On ordinal ranks of Baire class functions*, Fundamenta Mathematicae 247 (2019), 109–130; arXiv:1701.05649.
- [2] M. Elekes, V. Kiss, and Z. Vidnyánszky, *Ranks on the Baire class ξ functions*, Transactions of the American Mathematical Society 368 (2016), 8111–8143; arXiv:1406.5724.
- [3] A. S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics 156, Springer, 1995.